

Mathematical Foundations of Political Methodology

Robert Gulotty

University of Chicago

February 8, 2017

Day 2 : Conditional probability

- Conditional Probability
- Bayes Rule
- Independence
- Counting

Why study conditional probability?

Most of the relationships we are interested in as policy makers/analysts, are conditional ones. Knowing that an event has occurred, might lead us to reassess the probability of another event.

- What is the probability of Iran obtaining nuclear weapons given that it has a uranium enrichment program?
- What is the probability that we will be able to affect climate change if we adopt the climate change treaty?
- What is the probability of holding liberal political views given certain demographic characteristics?

Definition of Conditional probability

The *conditional probability* of an event A , given an event B is given by:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

for all $P(B) \neq 0$

We can rearrange this to form the “Multiplication Rule”:

$$P(A \cap B) = P(A|B) \times P(B)$$

Examples of Conditional Probability

Example 1:

- $F = \{\text{Myanmar sells oil to China}\}$
- $E = \{\text{Myanmar discovers oil}\}$
- If F occurs then E must occur, $P(E|F) = 1$

Example 2: Security Council voting on Syria strike

- $V = \{\text{Russia and China veto strike}\}$
- $A = \{\text{Strike approved by UN}\}$
- $P(A|V) = \frac{P(A \cap V)}{P(V)} = \frac{P(\emptyset)}{P(V)} = 0$

Axioms of Conditional probability

$P(A|B)$ is a probability, so it follows the axioms.

1 $P(A|B) \geq 0, \quad \forall A$

2 $P(\Omega|B) = 1.$

3 For all mutually exclusive events, $A_1, A_2, \dots A_n$

$$P(A_1 \cup A_2 \cup \dots \cup A_n|B) = P(A_1|B) + P(A_2|B) + \dots + P(A_n|B)$$

Conditional Probability Example

Basic claim: Foreign military threats (given certain other conditions) produce revolutions.

- Skocpol (1979)
- Let E denote the event of a revolution and
- Let F the presence of a foreign threat

The way to read Skocpol's hypothesis in probability language is:

$$P(E|F) \geq P(E|F^c)$$

Conditional Probability Example

How might we know whether $P(E|F) > P(E|F^c)$?

- $P(E|F) = \frac{P(E \cap F)}{P(F)}$
- $P(E|F^c) = \frac{P(E \cap F^c)}{P(F^c)}$
- So we need to know both the rate of Revolutions with foreign threats, the rate of Revolutions without foreign threats, and the probability of a foreign threat.

Revolutions

Skocpol's data:

	Revolution	No Revolution
Foreign Threat	France, Russia, China	
No Threat		England, Prussia, Japan

$P(A|B)$ and $P(B|A)$ are usually different

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

Example: Race and Welfare

‘I will go to the NAACP convention and tell the African-American community why they should demand paychecks instead of food stamps.’

-Newt Gingrich, January 5, 2012

Consider new enrollments to TANF

$P(\text{TANF Recipient} | \text{Black}) \approx 29\%$

$P(\text{TANF Recipient} | \text{White}) \approx 11\%$

$P(\text{Black} | \text{TANF Recipient}) \approx 26\%$

$P(\text{White} | \text{TANF Recipient}) \approx 47\%$

Confusing $P(A|B)$ and $P(B|A)$ leads to incorrect inference about welfare.

Law of Total Probability

For a set of mutually exclusive events, $E_1, E_2, E_3, \dots, E_n$, which together form the sample space Ω , the probability of any event A is:

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + \dots + P(A|E_n)P(E_n)$$

Why?

$$P(A) = P(A \cap \Omega)$$

$$P(A) = P(A \cap (E_1 \cup E_2 \cup \dots \cup E_n))$$

$$P(A) = P((A \cap E_1) \cup (A \cap E_2) \cup \dots \cup (A \cap E_n))$$

$$P(A) = P(A \cap E_1) + P(A \cap E_2) + \dots + P(A \cap E_n)$$

$$P(A) = P(A|E_1)P(E_1) + P(A|E_2)P(E_2) + \dots + P(A|E_n)P(E_n)$$

Example of Law of Total Probability

Say you decide to implement a Democracy Promotion Program (DPP) in some villages in Myanmar in the hopes of increasing voter turnout at the next election. What is $P(\text{vote})$ after DPP?

- Let $P(\text{vote} | DPP) = 0.6$
- Let $P(\text{vote} | \sim DPP) = 0.3$
- Let $P(DPP) = P(\sim DPP) = 0.5$ (Decided based on coin-flipping)
- $P(\text{vote}) = .6 * .5 + .3 * .5 = .45$

Example: Law of Total Probability

Infer $P(\text{vote})$ after mobilization campaign

- $P(\text{vote}|\text{mobilized}) = 0.75$
- $P(\text{vote}|\text{not mobilized}) = 0.25$
- $P(\text{mobilized}) = 0.6$; $P(\text{not mobilized}) = 0.4$
- What is $P(\text{vote})$?

Sample space (one person) =

{ (mobilized, vote), (mobilized, not vote), (not mobilized, vote) , (not mobilized, not vote) }

Mobilization partitions the space (mutually exclusive and exhaustive)

We can use the law of total probability

$$\begin{aligned}P(\text{vote}) &= P(\text{mob.}) \times P(\text{vote}|\text{mob.}) + P(\text{not mob}) \times P(\text{vote}|\text{not mob}) \\&= 0.6 \times 0.75 + 0.4 \times 0.25 \\&= 0.55\end{aligned}$$

Example Law of Total Probability

Mixture Models: flexible modeling strategy.

Two coins:

- Fair: $P(H) = 1/2$
- Biased $P(H) = 3/4$

Draw a coin from urn ($P(\text{fair}) = 1/2$) and then flip.

$P(H)$?

$$\Omega = \{(\text{fair}, H), (\text{fair}, T), (\text{bias}, H), (\text{bias}, T)\}$$

$$\begin{aligned} P(H) &= P(\text{fair}) \times P(H|\text{fair}) + P(\text{bias}) \times P(H|\text{bias}) \\ &= \frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{3}{4} \\ &= \frac{5}{8} \end{aligned}$$

Mixture of two coins

Bayes' Rule

- $P(B|A)$ may be easy to obtain
- $P(A|B)$ may be harder to determine
- Bayes' rule provides a method to move from $P(B|A)$ to $P(A|B)$.

Conditional Probability and Bayes' Rule

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

$$P(A \cap B) = P(B \cap A) = P(B|A)P(A)$$

Theorem

Bayes' Rule

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

General Statement of Bayes' Rule

Let the events $A_1 \dots A_k$ form a partition of the space S such that $P(A_j) > 0$ for $j = 1 \dots k$. And let B be an event such that $P(B) > 0$. Then for $i = 1 \dots k$

Theorem

Bayes' Rule

$$P(A_i|B) = \frac{P(A_i)P(B|A_i)}{\sum_{j=1}^k P(A_j)P(B|A_j)}$$

Bayes' Rule: Example

How do we identify racial groups from lists of names?

Census Bureau collects information on distribution of names by race.

For example, **Washington** is the “**blackest**” name in America.

- $P(\text{black}) = 0.126$.
- $P(\text{not black}) = 1 - P(\text{black}) = 0.874$.
- $P(\text{Washington} | \text{black}) = 0.00378$.
- $P(\text{Washington} | \text{not black}) = 0.000060615$.

$$\begin{aligned} P(\text{black} | \text{Wash}) &= \frac{P(\text{black})P(\text{Wash} | \text{black})}{P(\text{Wash})} \\ &= \frac{P(\text{black})P(\text{Wash} | \text{black})}{P(\text{black})P(\text{Wash} | \text{black}) + P(\text{nb})P(\text{Wash} | \text{nb})} \\ &= \frac{0.126 \times 0.00378}{0.126 \times 0.00378 + 0.874 \times 0.000060616} \\ &\approx 0.9 \end{aligned}$$

Monty Hall Problem

Suppose we have three doors. A, B, C .

Suppose you're on a game show, and you're given the choice of three doors: Behind one door is a car; behind the others, goats.

- A contestant guesses a door.
- The host opens a different door, revealing one of two goats. The contestant has option to switch.
- Should the contestant switch?

Contestant guesses A

$P(A) = 1/3 \rightsquigarrow$ chance of winning without switch

If C is revealed to not have a car:

Monty Hall Problem

$$\begin{aligned}P(B|C \text{ revealed}) &= \frac{P(B)P(C \text{ revealed}|B)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\&= \frac{1/3 \times 1}{1/3 + 1/3 \times 1/2} = \frac{1/3}{1/2} = \frac{2}{3}\end{aligned}$$

$$\begin{aligned}P(A|C \text{ revealed}) &= \frac{P(A)P(C \text{ revealed}|A)}{P(B)P(C \text{ revealed}|B) + P(A)P(C \text{ revealed}|A)} \\&= \frac{1/3 \times 1/2}{1/3 + 1/3 \times 1/2} = \frac{1}{3}\end{aligned}$$

Double chances of winning with switch

Independence and Information

Does one event provide **information** about another event?

Definition

If learning about B does not change the probability that A occurs, A is independent of B .

$$P(A|B) = P(A)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = P(A)$$

Independence

What if A is *Mutually Exclusive* of B ?

$$A \cap B = \emptyset$$

$$P(A|B) = \frac{P(\emptyset)}{P(B)} = 0 \neq P(A)$$

If two events occur with positive probability, then they cannot be mutually exclusive and independent.

Example Independence Relationship

Flip a fair coin twice.

E = first flip heads

F = second flip heads

$$\begin{aligned}P(E \cap F) &= P(\{(H, H), (H, T)\} \cap \{(H, H), (T, H)\}) \\&= P(\{(H, H)\})\end{aligned}$$

$$= \frac{1}{4}$$

$$P(E) = \frac{1}{2}$$

$$P(F) = \frac{1}{2}$$

$$P(E)P(F) = \frac{1}{2} \frac{1}{2} = \frac{1}{4} = P(E \cap F)$$

Independence: No Information

Suppose E and F are independent. Then,

$$\begin{aligned}P(E|F) &= \frac{P(E \cap F)}{P(F)} \\&= \frac{P(E)P(F)}{P(F)} \\&= P(E)\end{aligned}$$

Conditioning on the event F does not modify the probability of E .

No information about E in F

Independence and Compliments

Proposition

Suppose A and B are independent events. Then the events A and B^c are also independent.

Proof.

$$\begin{aligned}P(A \cap B^c) &= P(A) - P(A \cap B) \\&= P(A) - P(A)P(B) \\&= P(A)(1 - P(B)) \\&= P(A)P(B^c)\end{aligned}$$



Independence and Causal Inference

We want to know what the effect of some treatment is on some outcome of interest.

- Democracy promotion programs on democracy
- Cash transfers on education or health outcomes

In most studies, inference is made based on observations: observational studies. Units select into treatment, or are selected.

Example of Independence and Causal inference.

Example: democracy promotion programs in countries where we think they are likely to succeed

	Democracy	No Democracy
DPP	10	50
no DPP	2	13

$$P(\text{Democracy} | \text{DPP}) = \frac{(10/75)}{(60/75)} = 1/6 = .17$$

$$P(\text{Democracy} | \text{no DPP}) = \frac{(2/75)}{(15/75)} = 2/15 = .13$$

Example of Independence and Causal inference.

Example: democracy promotion programs in countries where we think they are likely to succeed

		Democracy	No Democracy
Good Gov	DPP	2	18
	no DPP	1	9
Bad Gov	DPP	8	32
	no DPP	1	4

$$P(\text{Democracy} | \text{DPP} \& \text{GG}) = \frac{(2/75)}{(20/75)} = .1$$

$$P(\text{Democracy} | \text{noDPP} \& \text{GG}) = \frac{(1/75)}{(10/75)} = .1$$

Conditional Independence

Definition

Let E_1 and E_2 be two events. We will say that the events are conditionally independent given E_3 if

$$P(E_1 \cap E_2 | E_3) = P(E_1 | E_3)P(E_2 | E_3)$$

Example: Independence and Causal Inference

Selection and Observational Studies

- We often want to infer the effect of some treatment
 - Incumbency on vote return
 - Democracy on war
- Observational studies: observe what we see to make inference
- Problem: units select into treatment
 - Simple example: enroll in job training if I think it will help
 - $P(\text{job}|\text{training in study}) \neq P(\text{job}|\text{forced training})$
- **Background characteristic**: difference between treatment and control groups
- **Experiments**: make background characteristics and treatment status independent

Conditional Probability

Definition

Let E_1 and E_2 be two events. We will say that the events are conditionally independent given E_3 if

$$P(E_1 \cap E_2 | E_3) = P(E_1 | E_3)P(E_2 | E_3)$$

Proposition

Suppose E_1 and E_2 and E_3 are events such that $P(E_1 \cap E_2) > 0$ and $P(E_2 \cap E_3) > 0$. Then E_1 and E_2 are conditionally independent given E_3 if and only if $P(E_1|E_2 \cap E_3) = P(E_1|E_3)$.

Proof.

Suppose E_1 and E_2 are conditionally independent given E_3 . Then

$$\begin{aligned}P(E_1 \cap E_2|E_3) &= \frac{P(E_1 \cap E_2 \cap E_3)}{P(E_3)} \\&= \frac{P(E_3)P(E_2|E_3)P(E_1|E_2 \cap E_3)}{P(E_3)} \\P(E_1|E_3)P(E_2|E_3) &= P(E_2|E_3)P(E_1|E_2 \cap E_3) \\P(E_1|E_3) &= P(E_1|E_2 \cap E_3)\end{aligned}$$



Proof.

Suppose $P(E_1|E_2 \cap E_3) = P(E_1|E_3)$

$$\begin{aligned} P(E_1 \cap E_2|E_3) &= P(E_2|E_3)P(E_1|E_2 \cap E_3) \\ &= P(E_2|E_3)P(E_1|E_3) \end{aligned}$$



Definition

Suppose we have a sequence of events $E_1, E_2, \dots, E_n$. We say the sequence of events is *mutually independent* if for each subset of the sequence, $E_{i_1}, E_{i_2}, \dots, E_{i_j}$

$$P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_j}) = \prod_{m=1}^j P(E_{i_m})$$

For a sequence to be independent, every subset is independent

Most Different research design (Method of Agreement)

Given a set of possible causes, A, B, C, D, and E, if we observe an outcome x after A, B and C in one case and after A, D, and E in another, then we have evidence that A causes x . This is sometimes called the method of elimination, where we rule out competing explanations.

- We must assume that the cause is deterministic.
- We must assume that one of these causes must be the cause (we aren't missing anything).
- We have ruled out B, C, D and E as necessary conditions.

Most Similar research design (Method of Difference)

Given the complete set of possible causes, A, B, and C, if we observe an outcome x after A, B and C and not after B and C alone, then we have evidence that A causes x .

- We must assume that the cause is deterministic.
- We must assume that all conditions other than A are exactly the same.
- We have established that A is a necessary condition.

A note on Mill's Method

“Nothing can be more ludicrous than the sort of parodies on experimental reasoning which one is accustomed to meet with, not in popular discussion only, but in grave treatises, when the affairs of nations are the theme. [...] “How can such or such causes have contributed to the prosperity of one country, when another has prospered without them?” Whoever makes use of an argument of this kind, not intending to deceive, should be sent back to learn the elements of some one of the more easy physical sciences.”

-John Stuart Mill. 1872. A System of Logic

A note on Mill's Method

John Stewart Mill rejected the use of his method for social science, particularly comparative and international relations, but they are still influential.